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## ACRYLIC

### GENERAL DESCRIPTION

Acrylics (Polymethyl-Methacrylate or PMMA) is an amorphous thermoplastic which is optically transparent, unaffected by moisture, and offers a high strength-to-weight ratio. Common trade names of acrylic include Plexiglas<sup>®</sup>, Lucite<sup>®</sup>, and Acrylite<sup>®</sup>.

### GENERAL PROPERTIES

Acrylics offer high light transmittance with a Refractive Index of 1.49 and can be easily heat-formed without loss of optical clarity. Prolonged exposure to moisture, or even total immersion in water, does not significantly effect the mechanical or optical properties of acrylic. Most commercial acrylics have been UV stabilized for good weatherability and resistance prolonged sunlight exposure.

Acrylics are unaffected by aqueous solutions of most laboratory chemicals, by detergents, cleaners, dilute inorganic acids, alkalies, and aliphatic hydrocarbons -- however, acrylics are NOT recommended for use with chlorinated or aromatic hydrocarbons, esters, or ketones.

Acrylics are easily sawed, drilled, milled, engraved, and finished with sharp carbide-tipped tools. Cut surfaces may be readily sanded and polished. They are also readily bend or thermoformed at low temperature and solvent bonding of properly fitting parts produces a strong, invisible joint. Acrylics are available in colorless clear as well as a wide variety of colors and tints. They are available in extruded and/or cast material in sheet, rod and tube forms as well as custom profiles.

### TYPICAL APPLICATIONS

Acrylic is an economical, general purpose material used in a wide variety of applications, including: • store fixtures and displays • lenses and lighting fixtures • light pipes • windows and skylights • sight gauges • furniture • outdoor signs • sculpture

### TYPICAL PROPERTIES of ACRYLIC PMMA

| ASTM or UL test | Property                                              | Acrylic       |
|-----------------|-------------------------------------------------------|---------------|
| <b>PHYSICAL</b> |                                                       |               |
| D792            | Density (lb/in <sup>3</sup> )<br>(g/cm <sup>3</sup> ) | 0.043<br>1.18 |
| D570            | Water Absorption, 24 hrs (%)                          | 0.3           |

|                   |                                                                                              |                   |
|-------------------|----------------------------------------------------------------------------------------------|-------------------|
|                   |                                                                                              |                   |
| <b>MECHANICAL</b> |                                                                                              |                   |
| D638              | Tensile Strength (psi)                                                                       | 8,000 - 11,000    |
| D638              | Tensile Modulus (psi)                                                                        | 350,000 - 500,000 |
| D638              | Tensile Elongation at Break (%)                                                              | 2                 |
| D790              | Flexural Strength (psi)                                                                      | 12,000 - 17,000   |
| D790              | Flexural Modulus (psi)                                                                       | 350,000 - 500,000 |
| D695              | Compressive Strength (psi)                                                                   | 11,000 - 19,000   |
| D695              | Compressive Modulus (psi)                                                                    | -                 |
| D785              | Hardness, Rockwell                                                                           | M80 - M100        |
| D256              | IZOD Notched Impact (ft-lb/in)                                                               | 0.3               |
| <b>THERMAL</b>    |                                                                                              |                   |
| D696              | Coefficient of Linear Thermal Expansion<br>( $\times 10^{-5}$ in./in./°F)                    | 5 - 9             |
| D648              | Heat Deflection Temp (°F / °C)<br>at 264 psi                                                 | 150-210 / 65-100  |
| D3418             | Melting Temp (°F / °C)                                                                       | 265-285 / 130-140 |
| -                 | Max Operating Temp (°F / °C)                                                                 | 150-200 / 65-93   |
| C177              | Thermal Conductivity<br>(BTU-in/ft <sup>2</sup> -hr-°F)<br>( $\times 10^{-4}$ cal/cm-sec-°C) | 3.9<br>1.2        |
| UL94              | Flammability Rating                                                                          | -                 |
| <b>ELECTRICAL</b> |                                                                                              |                   |
| D149              | Dielectric Strength (V/mil) short time, 1/8" thick                                           | 400               |
| D150              | Dielectric Constant at 60 Hz                                                                 | 4.0               |
| D150              | Dissipation Factor at 60 Hz                                                                  | 0.05              |
| <b>OPTICAL</b>    |                                                                                              |                   |
| -                 | Light Transmission, minimum (%)                                                              | 92                |
| -                 | Refractive Index                                                                             | 1.48-1.50         |

NOTE: The information contained herein are typical values intended for reference and comparison purposes only. They should NOT be used as a basis for design specifications or quality control. Contact us for manufacturers' complete material property datasheets. All values at 73°F (23°C) unless otherwise noted.

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